

Homework 1

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Due: 2026/09/30 10:59pm

1. First, illustrate the motivation behind introducing the concept of a random variable. Then, explain why a random variable, which is in fact a mapping (i.e., a function), is nevertheless called a random variable, even though the term itself suggests that it is a variable.
2. Show the proof of following results
 - (a) General Bayes' Rule (four cases).

	Y discrete	Y continuous
X discrete	$P(Y = y X = x) = \frac{P(X=x Y=y)P(Y=y)}{P(X=x)}$	$f_Y(y X = x) = \frac{P(X=x Y=y)f_Y(y)}{P(X=x)}$
X continuous	$P(Y = y X = x) = \frac{f_X(x Y=y)P(Y=y)}{f_X(x)}$	$f_{Y X}(y x) = \frac{f_{X Y}(x y)f_Y(y)}{f_X(x)}$

- (b) General LOTP (four cases).

	Y discrete	Y continuous
X discrete	$P(X = x) = \sum_y P(X = x Y = y)P(Y = y)$	$P(X = x) = \int_{-\infty}^{\infty} P(X = x Y = y)f_Y(y)dy$
X continuous	$f_X(x) = \sum_y f_X(x Y = y)P(Y = y)$	$f_X(x) = \int_{-\infty}^{\infty} f_{X Y}(x y)f_Y(y)dy$

- (c) Law of Total Variance. For any r.v.s X and Y ,

$$\text{Var}(Y) = E(\text{Var}(Y|X)) + \text{Var}(E(Y|X)).$$

- (d) The Linear Least Square Estimate (LLSE) of Y given X , denoted by $L[Y|X]$, is the linear function $a + bX$ that minimizes $E[(Y - a - bX)^2]$. In fact,

$$L[Y|X] = E(Y) + \frac{\text{Cov}(X, Y)}{\text{Var}(X)}(X - E(X))$$

- (e) Orthogonality Property of MMSE. For any function $\phi(\cdot)$, one has

$$E[(Y - E[Y|X])\phi(X)] = 0$$

Moreover, if the function $g(X)$ is such that

$$E[(Y - g(X))\phi(X)] = 0, \forall \phi(\cdot).$$

then $g(X) = E(Y|X)$

- (f) Let Z be a random variable with finite variance σ^2 . Then for any constant $a > 0$,

$$P(|Z - E[Z]| \geq a) \leq \frac{2\sigma^2}{\sigma^2 + a^2}.$$

- (g) General Hoeffding Inequality. Let the random variables $X_1, X_2, \dots, X_n$ be independent, with $a_k \leq X_k \leq b_k$ for each k , where a_k, b_k are constants. Let $S_n = \sum_{k=1}^n X_k$ and let $\mu = \mathbb{E}(S_n)$. Then for any $t \geq 0$,

$$\mathbb{P}(|S_n - \mu| \geq t) \leq 2e^{-\frac{2t^2}{\sum_{k=1}^n (b_k - a_k)^2}}.$$

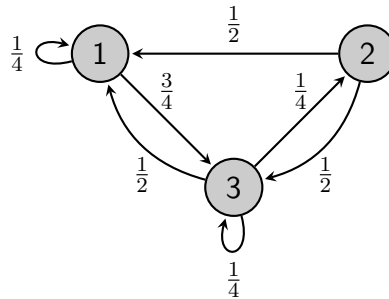
3. Let $X_1 \sim \text{Expo}(\lambda_1)$, $X_2 \sim \text{Expo}(\lambda_2)$ and $X_3 \sim \text{Expo}(\lambda_3)$ be independent. Find $E(X_1|X_1 > 2026)$, $E(X_1|X_1 < 2026)$ and $E(X_1 + X_2 + X_3|X_1 > 1997, X_2 > 2014, X_3 > 2026)$
4. Let Z_1, Z_2 be two *i.i.d.* random variables satisfying standard normal distributions, *i.e.*, $Z_1, Z_2 \sim \mathcal{N}(0, 1)$. Define

$$X = \sigma_X Z_1 + \mu_X;$$

$$Y = \sigma_Y \left(\rho Z_1 + \sqrt{1 - \rho^2} Z_2 \right) + \mu_Y,$$

where $\sigma_X > 0$, $\sigma_Y > 0$, $-1 < \rho < 1$.

- (a) Show that X and Y are bivariate normal.
- (b) Find the correlation coefficient between X and Y , *i.e.*, $\text{Corr}(X, Y)$.
- (c) Find the joint PDF of X and Y .
5. Given a time-homogeneous Markov chain with state-transition diagram shown as follows:



- (a) Find $P(X_5 = 3|X_4 = 2)$ and $P(X_4 = 1|X_3 = 2)$.
 - (b) Find the stationary distribution of the chain.
 - (c) Find the period of state 2 and the period of the chain.
 - (d) If $P(X_0 = 3) = \frac{1}{5}$, find $P(X_0 = 3, X_1 = 2, X_2 = 1)$.
 - (e) Find $E(X_5|X_3 = 1)$ and $\text{Var}(X_5|X_3 = 1)$
6. Given a coin with the probability p of landing heads. p is unknown and we need to estimate its value through data. In our data collection model, we have n independent tosses, result of each toss is either Head or Tail. Let X denote the number of heads in the total n tosses. Now we conduct experiments to collect data and find $X = k$. Then we need to find $\hat{p}$, the estimation of p .
- (a) Assume p is an unknown constant. Find $\hat{p}$ through the MLE (Maximum Likelihood Estimation) rule.
 - (b) Assume p is a random variable with a prior distribution $p \sim \text{Beta}(1, 1)$. Find $\hat{p}$ through the MAP (Maximum a Posterior Probability) rule.
 - (c) Assume p is a random variable with a prior distribution $p \sim \text{Unif}(0, 1)$. Find $\hat{p}$ through the MMSE (Minimal Mean Squared Error) rule.
 - (d) Assume p is a random variable with a prior distribution $p \sim \text{Unif}(0, 1)$. Find $\hat{p}$ through the LLSE (Linear Least Square Estimate) rule where

$$L[Y|X] = E(Y) + \frac{\text{Cov}(X, Y)}{\text{Var}(X)}(X - E(X)).$$

7. Let X, Y be jointly Gaussian random variables. Show the following equality with two different methods:

$$E[Y|X] = L[Y|X] = E(Y) + \frac{\text{Cov}(X, Y)}{\text{Var}(X)}(X - E(X)).$$

8. Find the expectation of the returned value of the following program code:

```
float sum=0.0;
while (sum<=1.0) {
    sum+= Unif(0,1);
}
return sum;
```

9. (Challenging) Prove Hoeffding Lemma. Let the random variable X satisfy $\mathbb{E}(X) = 0$ and $a \leq X \leq b$, where a and b are constants. Then for any $\lambda > 0$,

$$\mathbb{E}(e^{\lambda X}) \leq e^{\frac{1}{8}\lambda^2(b-a)^2}.$$

10. (Challenging) A resource-allocation system serves two users. Let x_1 and x_2 denote their service rates, measured in normalized units. Serving user 2 requires twice as much resource per unit of service as serving user 1. The feasible allocations are therefore

$$\mathcal{F} = \{(x_1, x_2) \in \mathbb{R}_{\geq 0}^2 : x_1 + 2x_2 \leq 12\}.$$

A system designer evaluates allocations using an aggregate utility function

$$W(x_1, x_2) = w_1 u_1(x_1) + w_2 u_2(x_2), \quad w_1, w_2 > 0.$$

Whenever logarithmic utility is used, restrict attention to strictly positive service rates. Throughout the problem, $\log$ denotes the natural logarithm.

Central question. How does the choice of utility determine what counts as a good allocation, and how can this objective be expressed as a reinforcement-learning reward?

- (a) What does a utility function represent? Explain what it means for a function u to represent a preference relation $\succeq$ over deterministic outcomes. Express your answer mathematically.

Suppose that both $u(x) = x$ and $v(x) = \log x$ represent the preference “more service is better.” Do they necessarily lead to the same allocation when the designer maximizes the sum of the users’ utilities? Explain why the ordering of an individual user’s outcomes does not, by itself, determine how outcomes should be compared across users.

- (b) What does concavity mean? Suppose that both users have the same twice-differentiable utility function u , with

$$u'(x) > 0, \quad u''(x) < 0.$$

Explain the meaning of each inequality. Sketch a representative utility curve and indicate how its slope changes as service increases.

Let $0 < a < b$. Prove that transferring a small amount of service from the better-served user to the worse-served user increases aggregate utility:

$$u(a + \delta) + u(b - \delta) > u(a) + u(b), \quad 0 < \delta < \frac{b - a}{2}.$$

Does this result imply that maximizing a sum of strictly concave utilities always gives equal service rates? Discuss the role of feasibility, using the resource constraint in this problem.

- (c) Which fairness criterion corresponds to logarithmic utility? Find the optimal allocation under each of the following objectives:

$$\text{Total service:} \quad \max_{(x_1, x_2) \in \mathcal{F}} (x_1 + x_2),$$

$$\text{Logarithmic utility:} \quad \max_{\substack{(x_1, x_2) \in \mathcal{F} \\ x_1, x_2 > 0}} (\log x_1 + \log x_2),$$

$$\text{Minimum service:} \quad \max_{(x_1, x_2) \in \mathcal{F}} \min\{x_1, x_2\}.$$

Compare the three allocations. Does logarithmic utility maximize total service, equalize service, or implement a different principle?

More generally, suppose that $\mathcal{C} \subseteq \mathbb{R}_{>0}^n$ is convex and that $\mathbf{x}^*$ maximizes

$$W(\mathbf{x}) = \sum_{i=1}^n \log x_i$$

over $\mathcal{C}$. Prove that, for every $\mathbf{y} \in \mathcal{C}$,

$$\sum_{i=1}^n \frac{y_i - x_i^*}{x_i^*} \leq 0.$$

Interpret this inequality in words and name the corresponding fairness criterion. Finally, find the optimal allocation when the designer uses

$$W(x_1, x_2) = 2 \log x_1 + \log x_2.$$

What does the coefficient 2 represent?

- (d) Concavity under uncertainty: fairness or risk aversion? Separately, consider a single user choosing between a certain payoff and a lottery:

Option A: $X = 5$ with certainty,

$$\text{Option B: } X = \begin{cases} 1, & \text{with probability } \frac{1}{2}, \\ 9, & \text{with probability } \frac{1}{2}. \end{cases}$$

Compare the options under the objectives $\mathbb{E}[X]$ and $\mathbb{E}[\log X]$. Find the **certainty equivalent** of Option B under logarithmic utility: the amount c satisfying

$$\log c = \mathbb{E}[\log X].$$

Use Jensen's inequality to explain the general relationship between concavity and risk aversion. Why is this interpretation different from the fairness interpretation in part (b)?

- (e) The von Neumann–Morgenstern expected utility theorem. State the main content of the **von Neumann–Morgenstern Expected Utility Theorem** for lotteries over a finite outcome set. Your statement should identify the four standard preference axioms, the expected-utility representation, and the transformations under which the representation is unique.

Does the theorem require utility to be concave or logarithmic? Does it determine a fair allocation across users?

Explain why applying an arbitrary strictly increasing transformation to utility can preserve preferences over deterministic outcomes while changing preferences over lotteries. Use part (d) as an illustration.

- (f) From utility to reinforcement-learning reward. Now suppose the system operates for T time steps. A policy π generates positive allocations

$$\mathbf{X}(t) = (X_1(t), X_2(t)) \in \mathcal{F}, \quad t = 0, \dots, T-1.$$

An undiscounted episodic RL agent maximizes

$$J(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} R_{t+1} \right].$$

State the **reward hypothesis** in your own words. Explain its connection to the expected utility theorem. In particular, does an expected-utility representation automatically provide a stationary Markov reward function on a prescribed state space?

Consider two possible fairness objectives:

$$J_{\text{instant}}(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} (\log X_1(t) + \log X_2(t)) \right],$$

and

$$J_{\text{average}}(\pi) = \mathbb{E}_\pi [\log \bar{X}_1 + \log \bar{X}_2], \quad \bar{X}_i = \frac{1}{T} \sum_{t=0}^{T-1} X_i(t).$$

For $T = 2$, compare these objectives for the following deterministic schedules:

$$\begin{aligned} \pi_A : & \quad (6, 3), (6, 3), \\ \pi_B : & \quad (10, 1), (2, 5). \end{aligned}$$

Are the objectives equivalent?

Construct an episodic reward whose expected cumulative sum equals $J_{\text{average}}(\pi)$. Explain what information may need to be added to the state.

- (g) Illustrate the connection between Von Neumann–Morgenstern Expected Utility Theorem and reward hypothesis in reinforcement learning.